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|--------|--------|
| Name:  | Date:  |
| Topic: | Class: |

| Main Ideas/Questions                                                                                                                                            | Notes/Examples                                                                                                                                                             |                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>WARM-UP</b>                                                                                                                                                  | <b>Directions:</b> Factor the following trinomials. Check for a GCF first.                                                                                                 |                                     |
|                                                                                                                                                                 | 1. $n^2 - 3n - 28$                                                                                                                                                         | 4. $4p^2 - 24p + 20$                |
| <b>FACTORING TRINOMIALS</b><br>of the form<br><br>$\underline{a}x^2 + bx + c$ | <b>When "a" cannot be factored out by GCF, we can possibly still factor the trinomial. The steps below show a method called "Slip and Slide" to factor the trinomials.</b> |                                     |
|                                                                                                                                                                 | <b>①</b> Fill in the first and last box with the first and last terms                                                                                                      |                                     |
|                                                                                                                                                                 | <b>②</b> Multiply "a" and "c" and find what factors will give you "b".                                                                                                     |                                     |
|                                                                                                                                                                 | <b>③</b> Fill in these two terms into the box.                                                                                                                             |                                     |
|                                                                                                                                                                 | <b>④</b> Determine what factors need to be on the outside of the box to give you the inside of the box. Distribute to check.                                               |                                     |
|                                                                                                                                                                 | <b>Example 1:</b> $2x^2 + 5x + 3$                                                                                                                                          | <b>Example 2:</b> $5m^2 - 17m + 6$  |
|                                                                                                                                                                 |                                                                                                                                                                            |                                     |
|                                                                                                                                                                 | <b>Example 3:</b> $6y^2 - 5y - 4$                                                                                                                                          | <b>Example 4:</b> $12c^2 + 11c - 5$ |
|                                                                                                                                                                 |                                                                                                                                                                            |                                     |

## YOU TRY!

1.  $3x^2 - 11x + 10$

2.  $8y^2 - 10y - 3$

3.  $9m^2 + 3m - 2$

4.  $10p^2 - p - 2$

5.  $4a^2 - 4a + 1$

6.  $6x^2 + x - 12$

## EXAMPLES WITH A GCF

**Directions:** Look for a GCF first, then factor the remaining trinomial.

7.  $10k^2 + 75k + 35$

8.  $12x^2 + 2x - 2$

9.  $15y^2 - 18y - 24$

10.  $32w^2 - 16w + 2$

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| Main Ideas/Questions                                                                                                                                        | Notes/Examples                                                                                                                                                      |                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WARM-UP</b>                                                                                                                                              | <b>Directions:</b> Factor the following trinomials. Check for a GCF first.                                                                                          |                                                                                                                                                                  |
|                                                                                                                                                             | 1. $x^2 + 14n + 45$<br>$(x+9)(x+5)$                                                                                                                                 | 2. $n^2 - 3n - 28$<br>$(n-7)(n+4)$                                                                                                                               |
|                                                                                                                                                             | 3. $2a^2 + 2a - 24$<br>$2(a^2 + a - 12)$<br>$2(a+6)(a-2)$                                                                                                           | 4. $4p^2 - 24p + 20$<br>$4(p^2 - 6p + 5)$<br>$4(p-5)(p-1)$                                                                                                       |
|                                                                                                                                                             | When "a" cannot be factored out by GCF, we can possibly still factor the trinomial. The steps below show a method called "Slip and Slide" to factor the trinomials. |                                                                                                                                                                  |
| <b>FACTORING TRINOMIALS</b><br>of the form<br><div style="border: 1px solid black; padding: 5px; display: inline-block;"> <math>ax^2 + bx + c</math> </div> | <b>1</b> "Slip" <u>a</u> to the end of the trinomial and multiply by <u>c</u> .                                                                                     |                                                                                                                                                                  |
|                                                                                                                                                             | <b>2</b> Factor this new basic trinomial.                                                                                                                           |                                                                                                                                                                  |
|                                                                                                                                                             | <b>3</b> Divide both your factors by the value you "slipped" over in Step 1.                                                                                        |                                                                                                                                                                  |
|                                                                                                                                                             | <b>4</b> Reduce fractions and "slide" any denominators up next to the variable. Distribute to check.                                                                |                                                                                                                                                                  |
|                                                                                                                                                             | <b>Example 1:</b> $2x^2 + 5x + 3$<br>$x^2 + 5x + 6$<br>$(x + \frac{3}{2})(x + \frac{3}{2})$<br>$(2x + 3)(x + 1)$                                                    | <b>Example 2:</b> $5m^2 - 17m + 6$<br>$m^2 - 17m + 30$<br>$(m - \frac{15}{5})(m - \frac{2}{5})$<br>$(m - 3)(5m - 2)$                                             |
|                                                                                                                                                             | <b>Example 3:</b> $6y^2 - 5y - 4$<br>$y^2 - 5y - 24$<br>$(y - \frac{8}{6})(y + \frac{3}{6})$<br>$(y - \frac{4}{3})(y + \frac{1}{2})$<br>$(3y - 4)(2y + 1)$          | <b>Example 4:</b> $12c^2 + 11c - 5$<br>$c^2 + 11c - 60$<br>$(c + \frac{15}{12})(c - \frac{4}{12})$<br>$(c + \frac{5}{4})(c - \frac{1}{3})$<br>$(4c + 5)(3c - 1)$ |

# YOU TRY!

1.  $3x^2 - 11x + 10$

$$x^2 - 11x + 30$$

$$(x - \frac{6}{3})(x - \frac{5}{3})$$

$$(x - 2)(3x - 5)$$

2.  $8y^2 - 10y - 3$

$$y^2 - 10y - 24$$

$$(y - \frac{6}{8})(y - \frac{4}{8})$$

$$(y - \frac{3}{4})(y - \frac{1}{2})$$

$$(4y - 3)(2y - 1)$$

3.  $9m^2 + 3m - 2$

$$m^2 + 3m - 18$$

$$(m + \frac{6}{9})(m - \frac{3}{9})$$

$$(m + \frac{2}{3})(m - \frac{1}{3})$$

$$(3m + 2)(3m - 1)$$

4.  $10p^2 - p - 2$

$$p^2 - p - 20$$

$$(p - \frac{5}{10})(p + \frac{4}{10})$$

$$(p - \frac{1}{2})(p + \frac{2}{5})$$

$$(2p - 1)(5p + 2)$$

5.  $4a^2 - 4a + 1$

$$a^2 - 4a + 4$$

$$(a - \frac{2}{4})(a - \frac{2}{4})$$

$$(a - \frac{1}{2})(a - \frac{1}{2})$$

$$(2a - 1)(2a - 1) \rightarrow (2a - 1)^2$$

6.  $6x^2 + x - 12$

$$x^2 + x - 12$$

$$(x + \frac{9}{6})(x - \frac{8}{6})$$

$$(x + \frac{3}{2})(x - \frac{4}{3})$$

$$(2x + 3)(3x - 4)$$

# EXAMPLES WITH A GCF

Directions: Look for a GCF first, then factor the remaining trinomial.

7.  $10k^2 + 75k + 35$

$$5(2k^2 + 15k + 7)$$

$$5(k^2 + 15k + 14)$$

$$5(k + \frac{14}{2})(k + \frac{1}{2})$$

$$5(k + 7)(2k + 1)$$

8.  $12x^2 + 2x - 2$

$$2(6x^2 + x - 1)$$

$$2(x^2 + x - 6)$$

$$2(x + \frac{3}{6})(x - \frac{2}{6})$$

$$2(x + \frac{1}{2})(x - \frac{1}{3})$$

$$2(2x + 1)(3x - 1)$$

9.  $15y^2 - 18y - 24$

$$3(5y^2 - 6y - 8)$$

$$3(y^2 - 6y - 40)$$

$$3(y - \frac{10}{5})(y + \frac{4}{5})$$

$$3(y - 2)(5y + 4)$$

10.  $32w^2 - 16w + 2$

$$2(16w^2 - 8w + 1)$$

$$2(w^2 - 8w + 16)$$

$$2(w - \frac{4}{16})(w - \frac{4}{16})$$

$$2(w - \frac{1}{4})(w - \frac{1}{4})$$

$$2(4w - 1)(4w - 1)$$

$$2(4w - 1)^2$$